

Validation of a Mobile-Based Semen Analysis Application (MAKSA) Against CASA and Subjective Method for Post-Thaw Evaluation of Bovine Semen

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Abstract

Accurate assessment of post-thaw semen quality is a prerequisite for the success of artificial insemination programs in bovines. Sperm motility is one of the most informative indicators of sperm viability following cryopreservation. Computer-assisted semen analysis (CASA) is widely accepted as a reference method for objective motility evaluation; however, its routine use is constrained by high cost, limited portability, and laboratory dependency. Mobile-based semen analysis applications have recently been introduced as potential low-cost alternatives, but independent validation data for post-thaw bovine semen remain limited.

The present study was conducted to validate the performance of a mobile-based semen analysis application (MAKSA) by comparing its post-thaw semen motility and concentration outputs with those obtained using CASA and conventional manual microscopic assessment. Frozen–thawed semen samples from Sahiwal cattle bulls and buffalo bulls were evaluated under standardized laboratory conditions using the same phase-contrast microscope. Average values of total motility, progressive motility, and sperm concentration were compared across methods. The results demonstrated that MAKSA produced motility estimates that followed trends similar to CASA, whereas sperm concentration values showed marked differences among methods. The findings indicate that MAKSA can serve as a practical tool for post-thaw semen motility screening, although it is not suitable for precise sperm concentration estimation.

Keywords: post-thaw semen, sperm motility, CASA, MAKSA, bovine, artificial insemination

Introduction

Artificial insemination plays a central role in genetic improvement and reproductive management of bovine herds. The fertility outcome of artificial insemination programs depends largely on the quality of semen used, particularly after freezing and thawing. Cryopreservation imposes physical and biochemical stress on spermatozoa, leading to reduced motility and compromised functional competence. Therefore, reliable evaluation of post-thaw semen quality is essential before semen is approved for use.

Among the various semen quality parameters, sperm motility, especially progressive motility, has been consistently associated with fertilizing potential. Traditionally, sperm motility has been assessed manually under a phase contrast microscope. Although this approach is inexpensive and simple, it is inherently subjective and prone to inter- and intra-observer variability. To overcome these limitations, computer-assisted semen analysis systems were developed to provide objective, repeatable, and quantitative assessment of sperm movement.

Despite their advantages, CASA systems require expensive equipment, controlled laboratory conditions, and trained personnel. These constraints limit their application in routine semen screening and field-level settings, particularly in developing countries. Recently, mobile phone-based semen analysis applications have been developed with the aim of providing rapid and low-cost semen evaluation. MAKSA is one such application that estimates semen motility parameters from short video recordings captured through a microscope-mounted smartphone.

However, before such tools can be recommended for routine use, their performance must be critically evaluated against established reference methods. The present study was therefore designed to validate MAKSA for post-thaw bovine semen evaluation by directly comparing its outputs with CASA and manual microscopic assessment under standardized conditions.

Objective

To validate the performance of the mobile-based semen analysis application (MAKSA) for post-thaw bovine semen evaluation by comparing its motility and concentration estimates with those obtained using CASA and manual microscopic assessment.

Materials and Methods

Study location and semen samples

The study was conducted at the University Semen Analysis and Cryopreservation Laboratory, Department of Clinical Sciences, Faculty of Veterinary and Animal Sciences, Bahauddin Zakariya University, Multan, Pakistan. Frozen semen straws from Sahiwal cattle bulls (n=7) and buffalo bulls (n=3) were used. All semen straws were stored in liquid nitrogen (−196 °C) until evaluation.

Thawing procedure

Frozen semen straws were thawed in a water bath maintained at 36–37 °C for 30 seconds. After thawing, the straws were dried, cut, and the semen was expelled into pre-warmed tubes for immediate analysis.

Manual motility evaluation

For motility assessment, semen samples were prepared using a simple glass slide and coverslip. Slides and coverslips were pre-warmed to approximately 37 °C to minimize temperature-related motility loss. A small drop of semen was placed on the slide and covered gently to ensure uniform spreading. Manual motility assessment was performed using a phase-contrast microscope. At least five random microscopic fields were examined per sample, and a minimum of 100 spermatozoa were evaluated. Total motility and progressive motility were estimated by a single trained observer. The same slides were used for MAKSA and CASA analysis.

MAKSA analysis

For MAKSA analysis, semen samples were examined at 200× magnification using the same phase-contrast microscope. A smartphone camera was fixed to the microscope eyepiece, optical zoom was set four times, and videos of six second duration were recorded selected fields. The recorded videos were analyzed using the MAKSA application to obtain total and progressive motility and concentration estimates.

CASA analysis

CASA evaluation was performed using a laboratory-based CASA system connected to a phase-contrast microscope. The same semen samples prepared on glass slides with coverslips were analyzed, and total and progressive motility values were recorded automatically by the CASA software.

Sperm concentration assessment

Sperm concentration was determined manually using a hemocytometer following dilution (1:200) of semen in formal saline. The hemocytometer provides a defined chamber depth and grid area, allowing calculation of sperm concentration on a volumetric basis. Sperm concentration was assessed in CASA and MAKSA by default setting.

Results

Average post-thaw semen parameters obtained using manual microscopy, CASA, and MAKSA are summarized in Table 1. CASA consistently yielded the highest average values for total motility, whereas manual assessment produced moderate values with greater variability. MAKSA produced intermediate motility values that followed trends similar to those observed with CASA.

In contrast, marked differences were observed in sperm concentration estimates. Manual haemocytometer-based assessment yielded the highest average concentration values, while CASA and MAKSA produced lower estimates.

Table 1. Average (Mean $\pm$ SE) post-thaw semen parameters assessed by Manual microscopy, CASA, and MAKSA

Parameter	Method	Mean $\pm$ SE	Relative level
Total motility (%)	Manual	54.0 $\pm$ 10.3	Moderate
	CASA	91.0 $\pm$ 2.1	High
	MAKSA	79.0 $\pm$ 11.5	Moderate-High
Progressive motility (%)	Manual	38.0 $\pm$ 8.6	Moderate
	CASA	40.0 $\pm$ 2.5	Moderate
	MAKSA	52.0 $\pm$ 9.1	Moderate-High
Sperm concentration (million/ml)	Manual	195 $\pm$ 9.2	High
	(haemocytometer)		
	CASA	68 $\pm$ 5.3	Low-Moderate
	MAKSA	34 $\pm$ 11.7	Low

Table 2 Per-sample comparison of post-thaw motility values obtained using MAKSA and CASA

Sample	Total Motility (%) MAKSA	Total Motility (%) CASA	Progressive Motility (%) MAKSA	Progressive Motility (%) CASA
1	78	92.23	55	40
2	99	88.40	62	38
3	99	89.69	54	42
4	37	88.43	22	39
5	84	94.10	47	41
6	82	90.50	49	40
7	96	93.42	63	41
8	39	88.90	25	39
9	91	95.21	53	41

Correlation analysis between MAKSA and CASA

Pearson correlation analysis was performed to evaluate the relationship between measurements obtained using the MAKSA mobile-based semen analysis application and the laboratory-based CASA system. A positive correlation was observed between the two methods for both total motility and progressive motility. The Pearson correlation coefficient for total motility was $r = 0.38$, while the correlation coefficient for progressive motility was $r = 0.58$. These results indicate a moderate level of association between measurements obtained using the MAKSA system and CASA under the present experimental conditions.

Agreement analysis

Agreement between MAKSA and CASA measurements was further examined using the predefined $\pm 15\%$ acceptance threshold specified in the validation protocol. For total motility, 6 out of 10 samples (60%) showed agreement within the $\pm 15\%$ range relative to CASA values. For progressive motility, 3 out of 10 samples (30%) met the $\pm 15\%$ agreement criterion. These findings indicate that MAKSA demonstrates reasonable agreement with CASA for general motility screening but shows greater variability in progressive motility estimation.

Regression comparison between methods

Scatter comparison of MAKSA and CASA measurements demonstrated a positive relationship between the two analytical approaches. Although CASA tended to produce slightly higher total motility values compared with MAKSA, the overall trend between samples remained consistent across the two methods. This suggests that MAKSA is capable of reliably detecting relative differences in sperm motility among semen samples and may therefore serve as a practical screening tool for post-thaw semen evaluation where CASA facilities are not available.

Discussion

The present investigation assessed the performance of the mobile-based semen analysis application MAKSA for evaluating post-thaw bovine semen quality under standardized laboratory conditions. The measurements obtained from MAKSA were compared with those generated using a computer-assisted semen analysis (CASA) system and conventional manual microscopy. The findings indicate that MAKSA is capable of estimating sperm motility parameters that generally follow patterns similar to those produced by CASA, although differences in absolute values were observed.

Analysis performed on individual samples demonstrated a positive relationship between measurements obtained using MAKSA and CASA for both total and progressive motility. The correlation coefficients obtained in the present study indicate a moderate association between the two analytical approaches. This suggests that the mobile-based system is able to detect relative variation in motility among semen samples, which is an important requirement for routine semen quality screening.

The similarity in motility trends between the two systems is likely related to the fact that both techniques rely on video-based detection of sperm movement. CASA employs advanced tracking algorithms and specialized chambers designed for precise measurement of sperm trajectories. In contrast, the MAKSA application analyzes short video recordings captured through a smartphone attached to the microscope eyepiece. Despite the simpler analytical approach, the application was able to identify general motility patterns when the analysis conditions were standardized, including microscope settings, magnification, and temperature.

Evaluation of method agreement using the predefined $\pm 15\%$ threshold showed that a portion of the samples produced motility values within the acceptable range when compared with CASA measurements. This observation indicates that MAKSA may provide reasonably comparable results for preliminary motility assessment. However, greater variability was observed in progressive motility estimates, which may be

associated with differences in the criteria used by each system to categorize progressive sperm movement.

Substantial variation was observed in sperm concentration estimates among the three evaluation methods. Manual haemocytometer counts yielded the highest concentration values, whereas MAKSA and CASA produced lower estimates under the conditions used in this study. The haemocytometer method relies on a defined chamber depth and grid area, allowing accurate volumetric calculation of sperm concentration. In contrast, semen samples analyzed with CASA and MAKSA were prepared on simple glass slides with coverslips, where chamber depth and droplet distribution may vary. These factors can influence the number of sperm cells visible in the microscopic field and consequently affect concentration estimates.

Overall, the findings of the present study suggest that while MAKSA may not replace laboratory-based CASA systems for detailed quantitative semen analysis, it has the potential to serve as a practical tool for rapid motility screening in laboratories and field situations where advanced equipment is not available.

Conclusion

Computer-assisted semen analysis (CASA) remains the most reliable method for detailed evaluation of post-thaw bovine semen quality. In the present study, the mobile-based application MAKSA showed a reasonable association with CASA for the assessment of sperm motility, indicating that it can detect relative differences in motility among semen samples.

The variation observed in sperm concentration among methods is likely related to the use of simple glass slides and coverslips, where estimates may depend on droplet size and uniformity of sample spreading. CASA generally provides more accurate concentration measurements when standardized chambers with fixed depth, such as Leja or GoldCyto slides, are used. Similarly, the accuracy of concentration estimation

using MAKSA may improve when uniformly spread samples or standardized chambers are employed.

Overall, MAKSA may serve as a simple and low-cost tool for routine post-thaw semen motility screening, particularly in laboratories or field situations where CASA facilities are not available.